

DNA METHYLATION PATTERNS AND SECRETORY IMMUNOGLOBULIN A (SIgA) LEVELS IN COLORECTAL CANCER PROGRESSION: AN EPIGENETIC APPROACH FROM ULCERATIVE COLITIS

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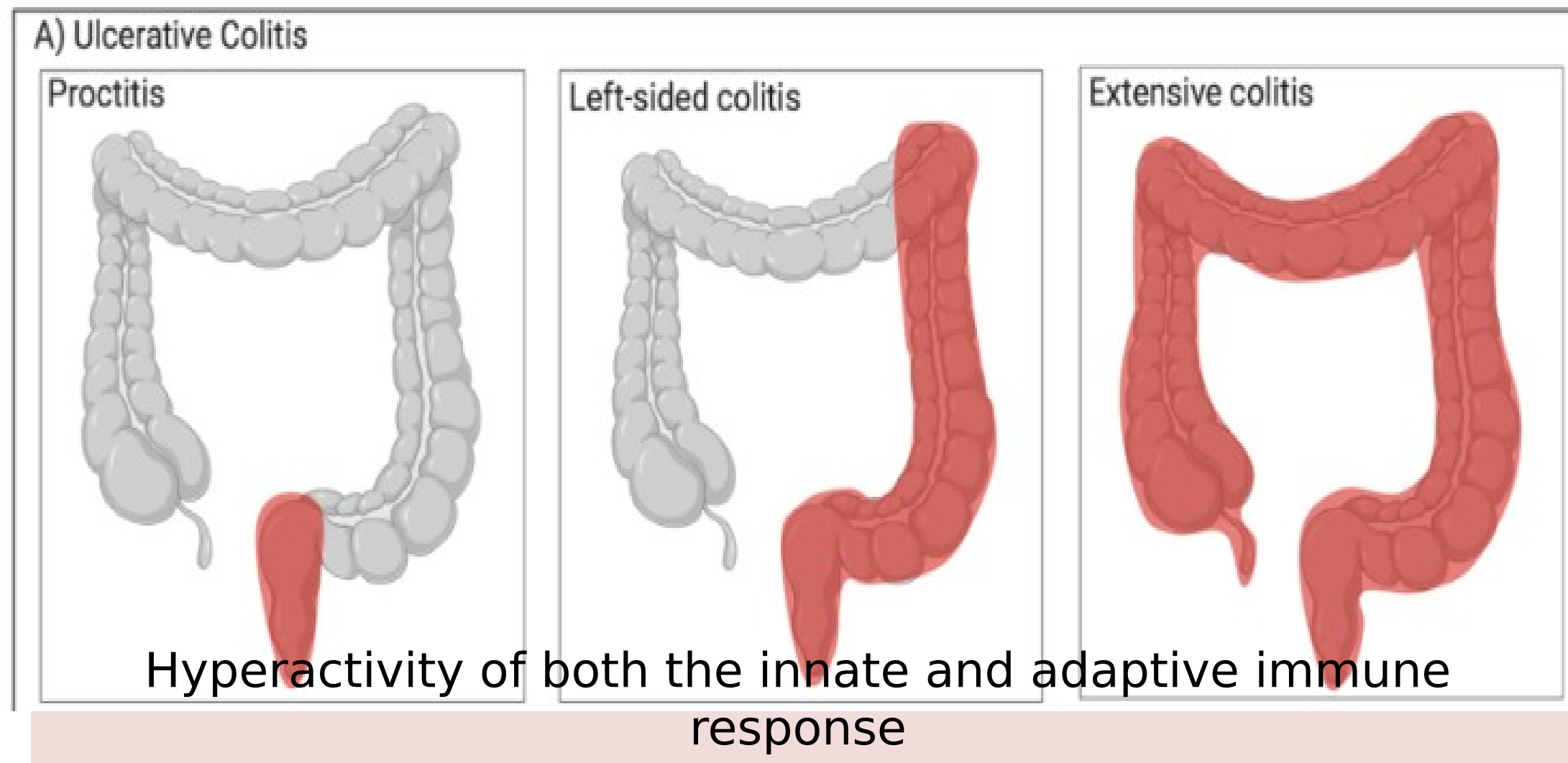
Universidad El Bosque – InmuBo - Cancer Research Line
2025

PCI-2024-0016

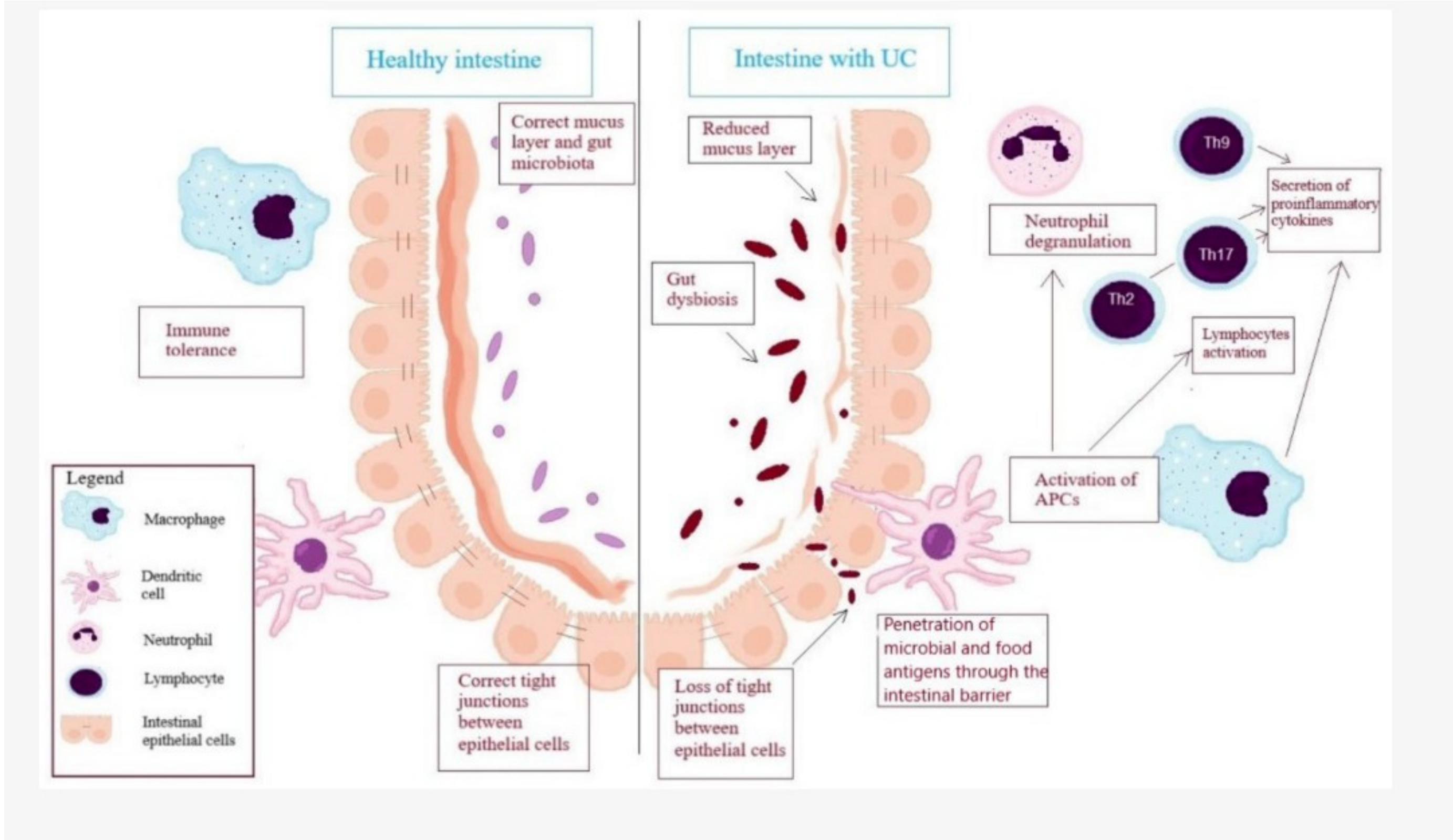


Ulcerative Colitis

Continuous inflammation of the mucosa
Begins in the rectum – extends through the colon

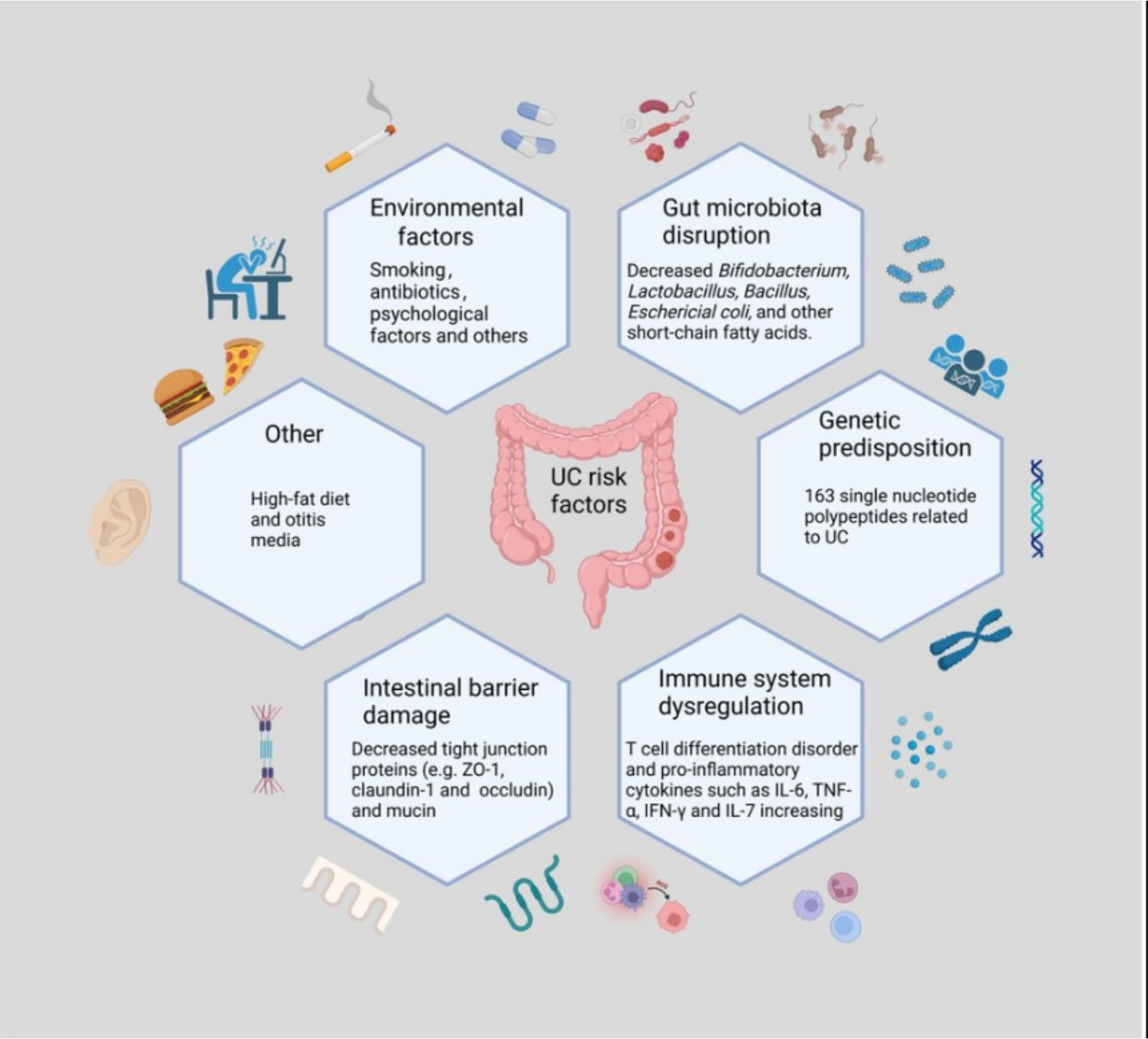


Pathophysiology



Uncontrolled cell proliferation favors tumor formation

Risk Factors in Ulcerative Colitis



European
Crohn's and Colitis
Organisation

Endoscopic surveillance programs

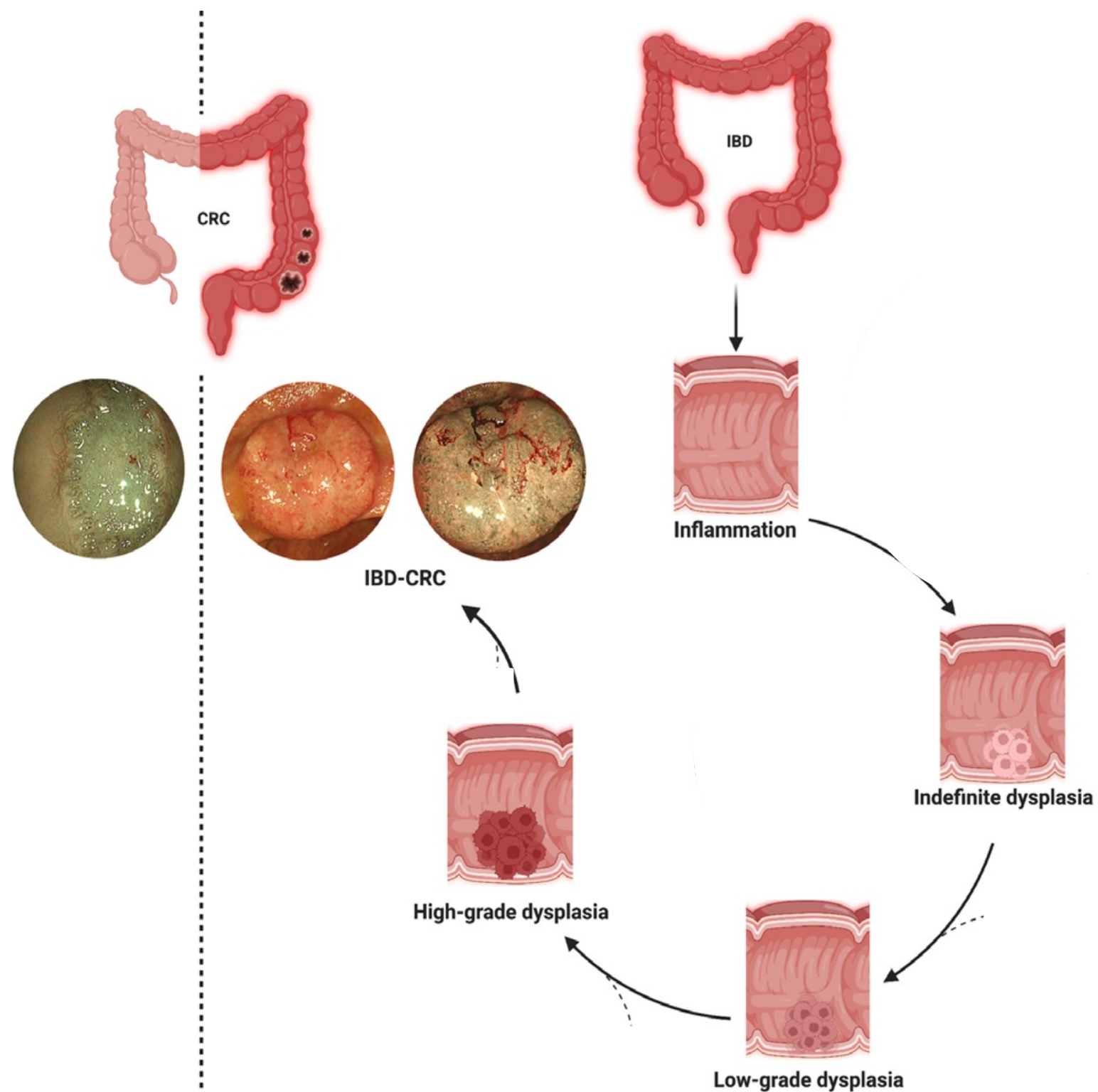
- 8 years from symptom onset in patients with IBD

Personalized follow-up strategy

Risk stratification
Early detection
Precancerous lesions

Kaser A, Zeissig S, Blumberg RS. Inflammatory bowel disease. Annu Rev Immunol. 2010;28(1):573–621. Available from: <http://dx.doi.org/10.1146/annurev-immunol-030409-1012251>.
1.Huang C, Hao W, Wang X, Zhou R, Lin Q. Probiotics for the treatment of ulcerative colitis: a review of experimental research from 2018 to 2022. Frontiers in Microbiology [Internet]. 2023 Jul 6;14. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10358780/>
Yeshi K, Ruscher R, Hunter L, Daly NL, Loukas A, Wangchuk P. Revisiting inflammatory bowel disease: Pathology, treatments, challenges and emerging therapeutics including drug leads from natural products. J Clin Med . 2020;9(5):1273.





Epithelial dysplasia, which can be high-grade, is a stage preceding the development of colorectal cancer, and its presence in biopsies predicts the coexistence of a carcinoma.

Cancer Associated with Colitis

UC associated with CRC is characterized by the progressive accumulation of genetic and epigenetic alterations and dysregulation of the mucosal immune system

Hypomethylated genes in severe UC

IL10, SIGLEC5, CD86, CLMP, NLRP3 and NLRC4

(anti-inflammatory activity)

Taman H, et al 2021

Hypermethylated genes in UC

Inverse correlation: methylation and gene expression

ROR1, GXYLT2, FOXA2

RARB associated with colorectal adenocarcinoma

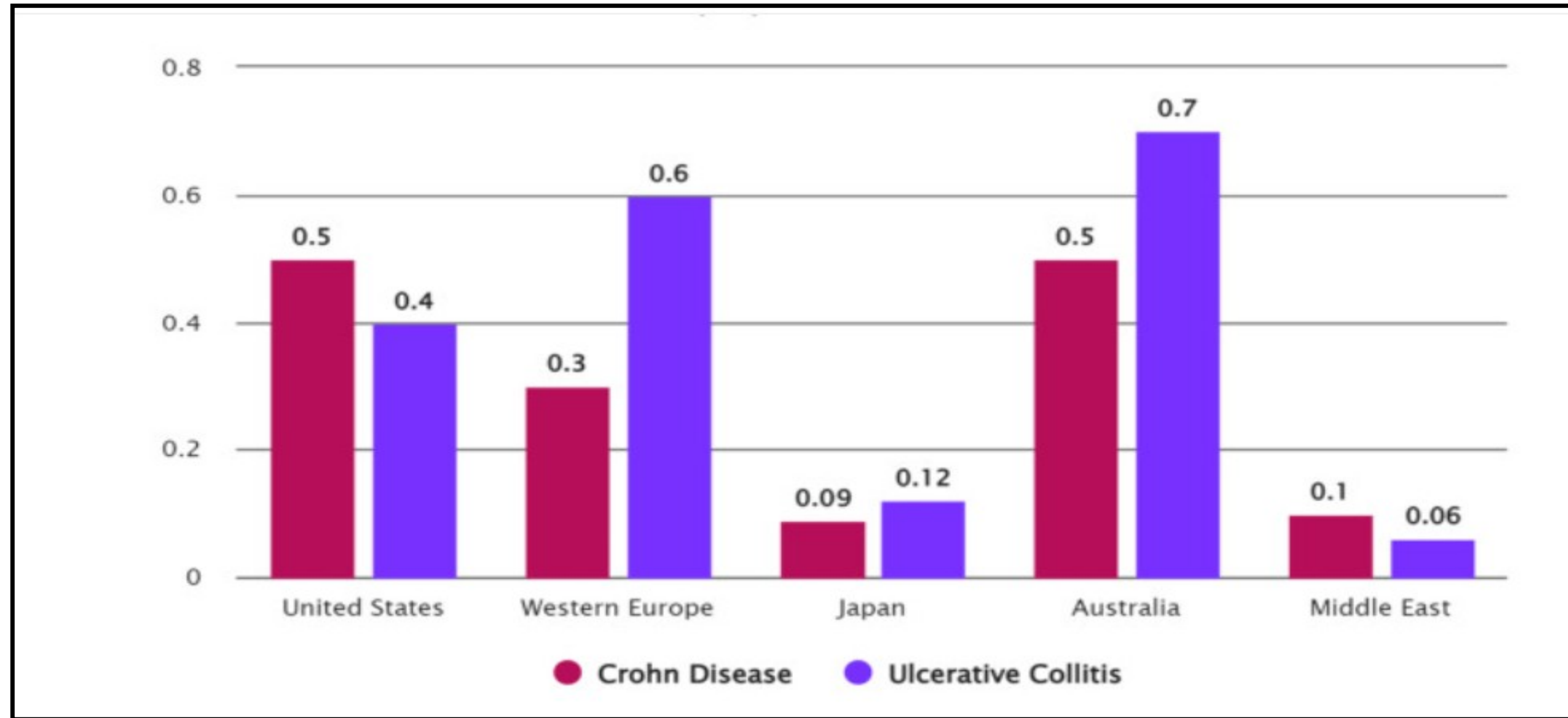
Barnicle A, et al 2017

Chronic inflammation and the associated risk become more relevant when considering the CRC-colitis association

T. Barnicle A, Seoighe C, Greally JM, Golden A, Egan LJ. Inflammation-associated DNA methylation patterns in epithelium of ulcerative colitis. *Epigenetics*. 2017 Aug 3;12(8):591–606.11,Nardone OM, Zammarchi I, Santacroce G, Ghosh S, Iacucci M. Inflammation-Driven Colorectal Cancer Associated with Colitis [Internet]. *Encyclopedia.pub*. Olga Maria Nardone; 2023 [cited 2025 Sep 20]. Available from: <https://encyclopedia.pub/entry/44048>

Incidence and Prevalence Inflammatory Bowel Disease

UC has become a global health challenge

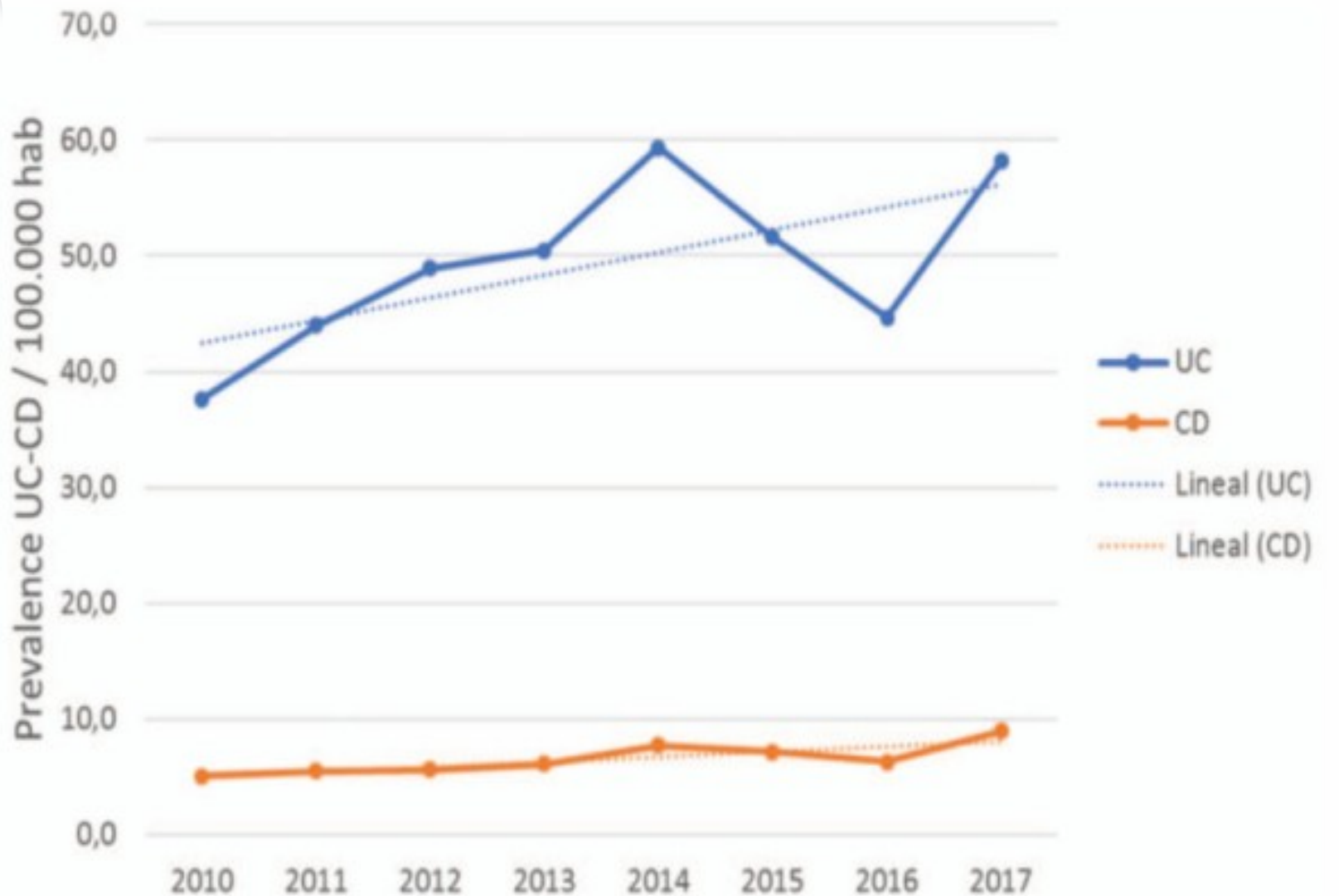
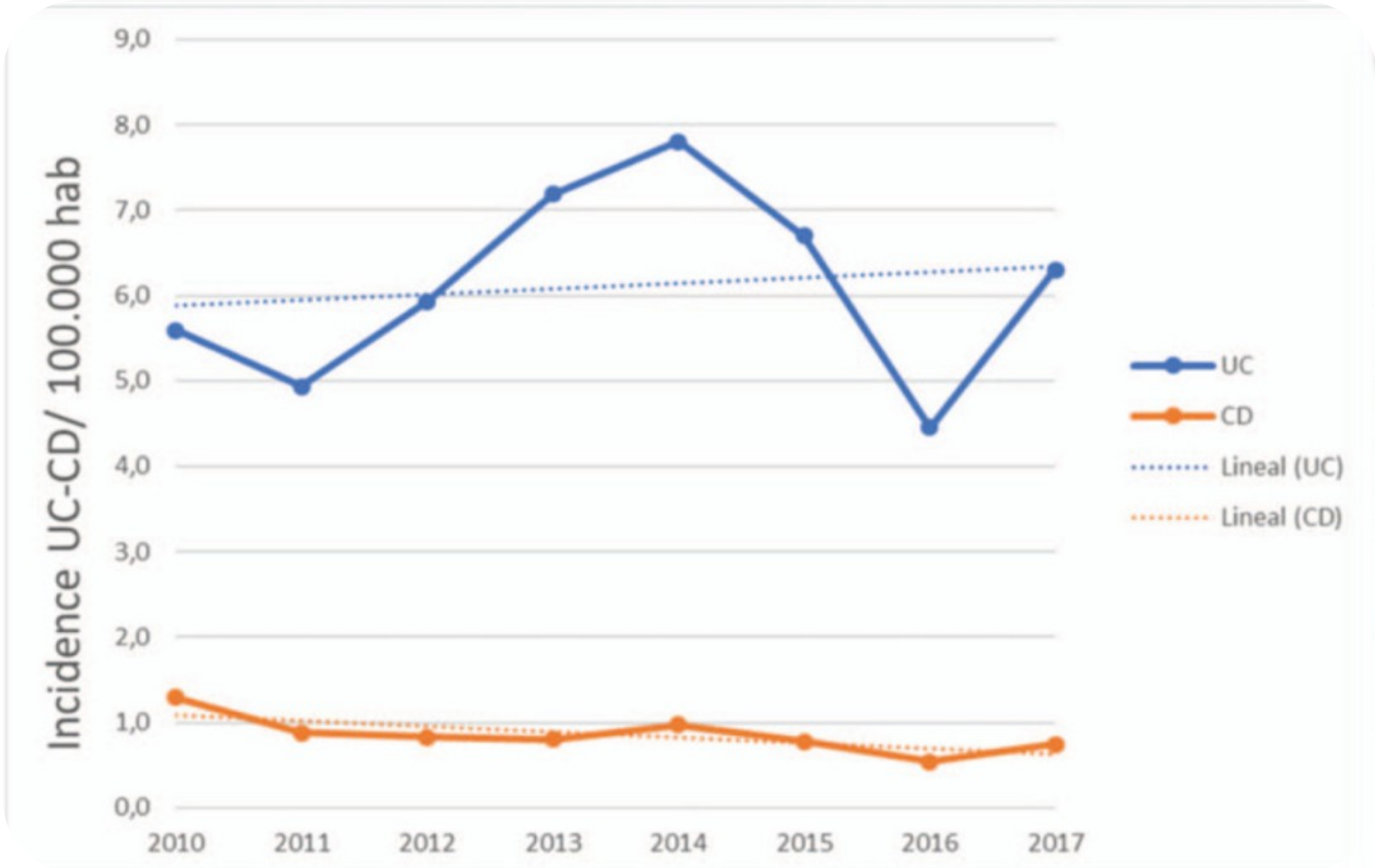


2000

2017

Incidence and Prevalence - Inflammatory Bowel Disease in Colombia

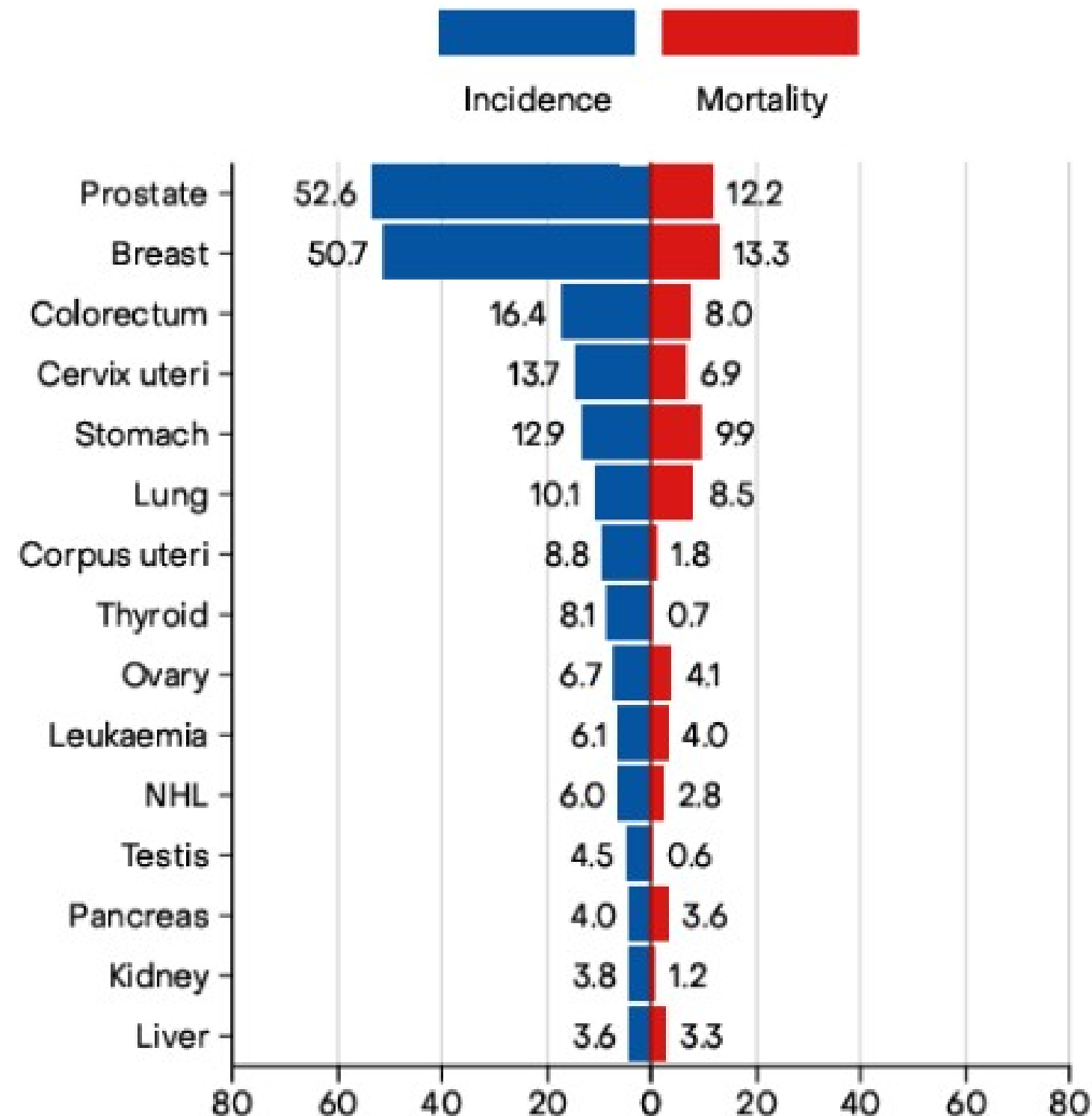
Suggest a growing burden of UC in Colombia



UC, along with its potential to progress to a neoplasm despite preventive treatments, represents a challenge for the scientific community

Mortality in Colombia

Chronic inflammation becomes critically important in the context of colorectal cancer (CRC) risk



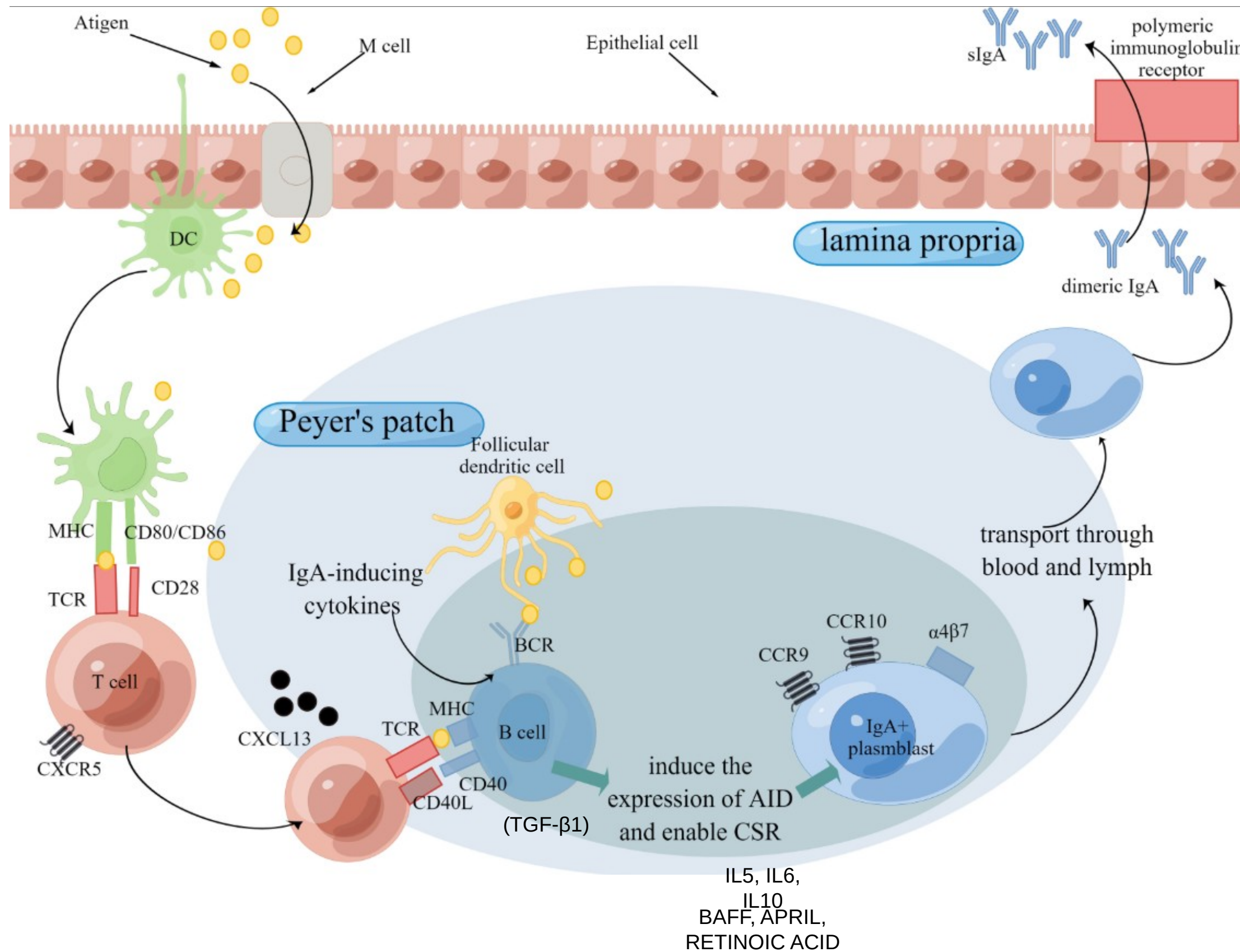
In Colombia, CRC is the third most common cancer

Exposure to risk factors:

- Late detection due to lack of screening and prevention programs, and reduced access to adequate treatment

Link

Persistent inflammation - malignant transformation.



Role of SIgA

- Limits systemic exposure to commensal microbes
- Modulates inflammation
- Shapes the adaptive immune response

Extensively studied in healthy conditions

Its role in IBD remains largely unknown

SIgA levels reported in serum

Increased serum SIgA levels may reflect subclinical intestinal involvement in axial and peripheral non-radiographic spondyloarthritis



Serum SIgA levels reported: Arias, et al 2021
Patients with SpA: 19.4 µg/ml (mean).
Healthy controls (HV): 9.4 µg/ml ($p < 0.0001$).
Patients with IBD: also showed higher levels than HV ($p = 0.004$), but lower than those with SpA ($p = 0.0476$)

Serum SIgA as a biomarker of subclinical intestinal inflammation in axial spondyloarthritis



Serum SIgA levels reported: Li L, et al. 2025
Patients with axSpA: serum SIgA levels significantly higher than healthy controls (median 16.5 µg/ml vs. 9.4 µg/ml; $p < 0.0001$).

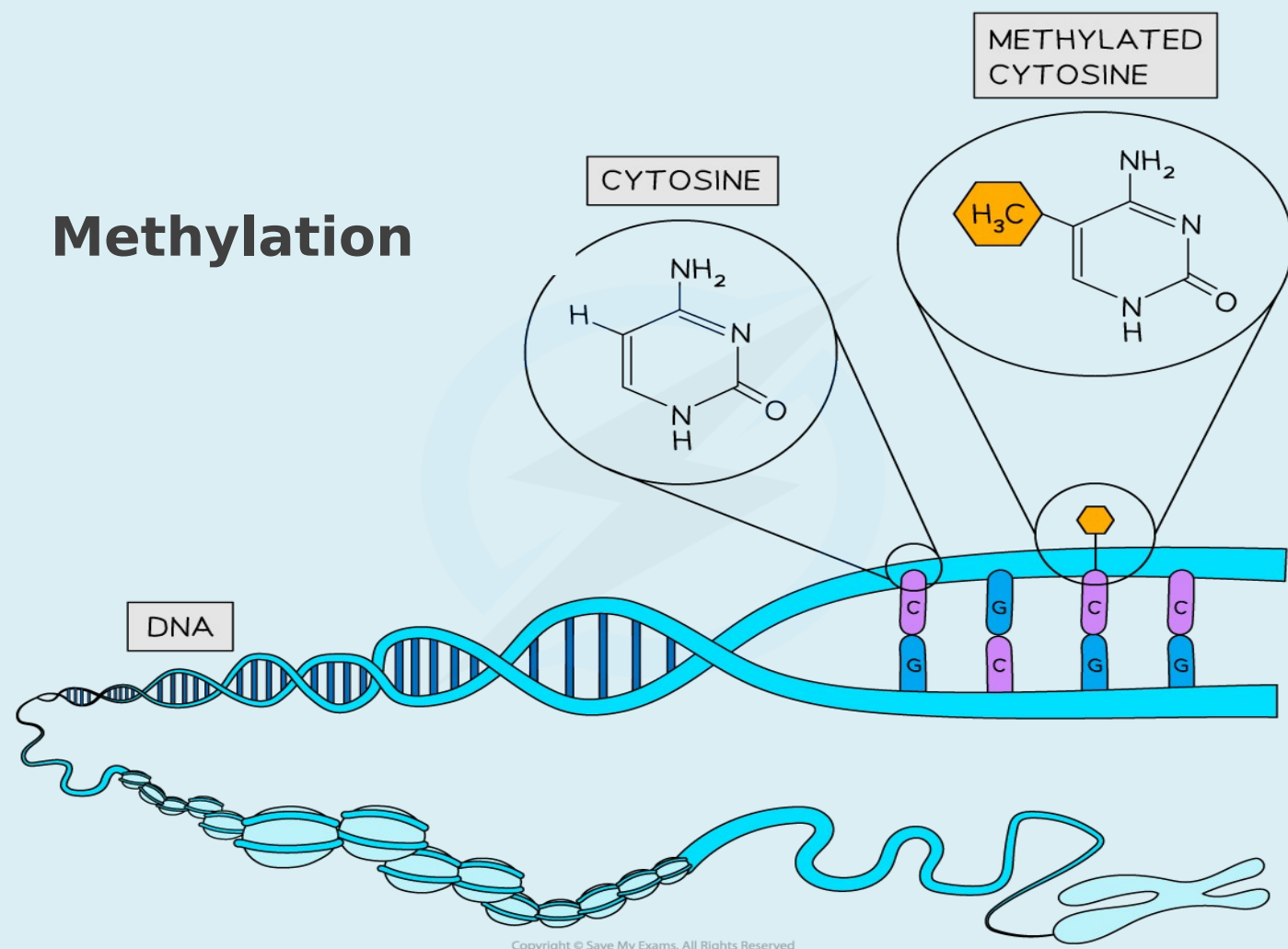
Elevated serum SIgA levels in axial spondyloarthritis: a possible biomarker reflecting subclinical intestinal inflammation



Serum SIgA levels reported: Cyfers H, et al 2017
Patients with axSpA showed serum SIgA levels significantly higher than controls (median 17.8 µg/ml vs. 9.4 µg/ml; $p < 0.0001$).

SIgA is key to reducing the inflammation-mediated response

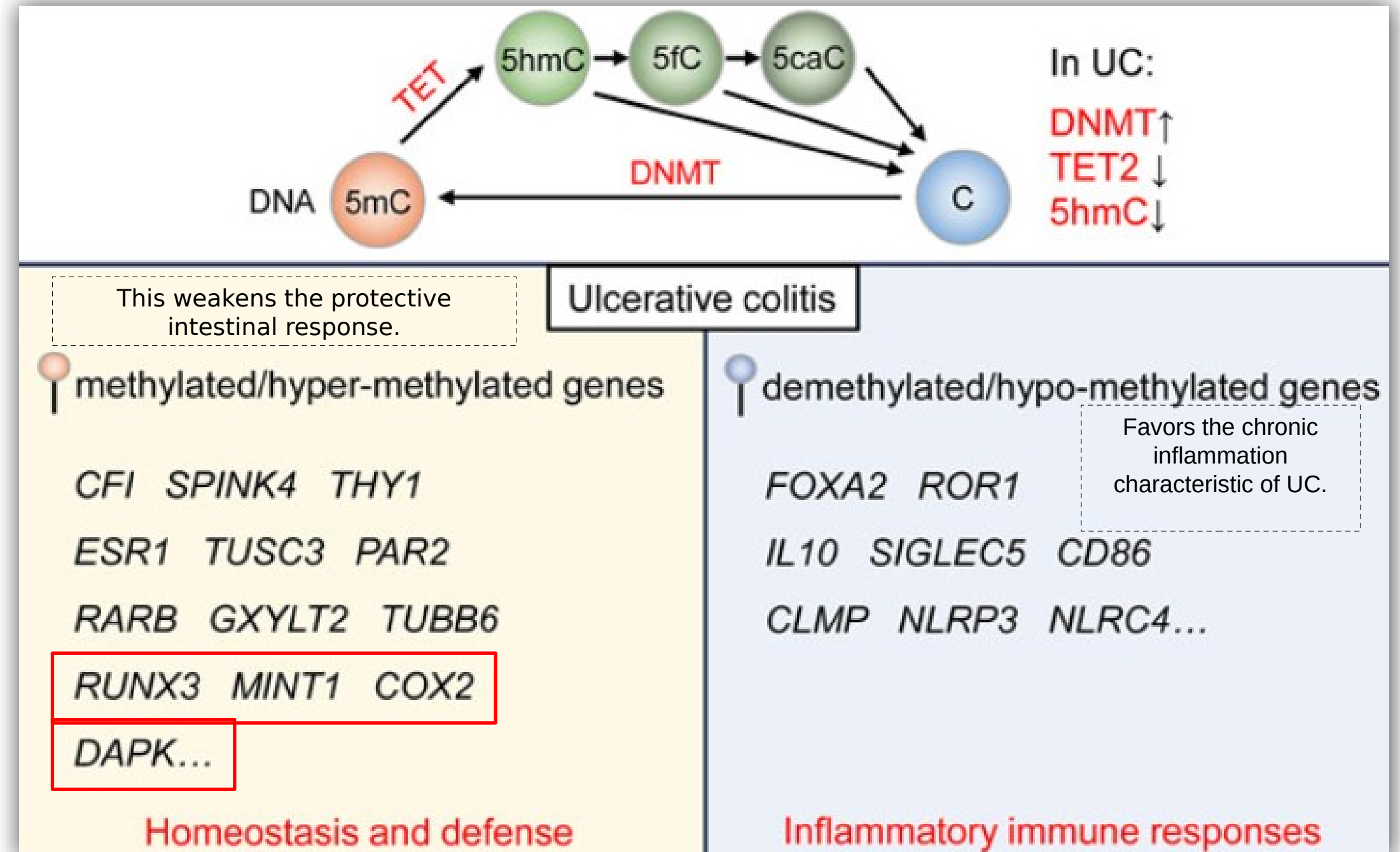
Methylation



Key mechanism

Regulates gene expression
Maintains genome stability

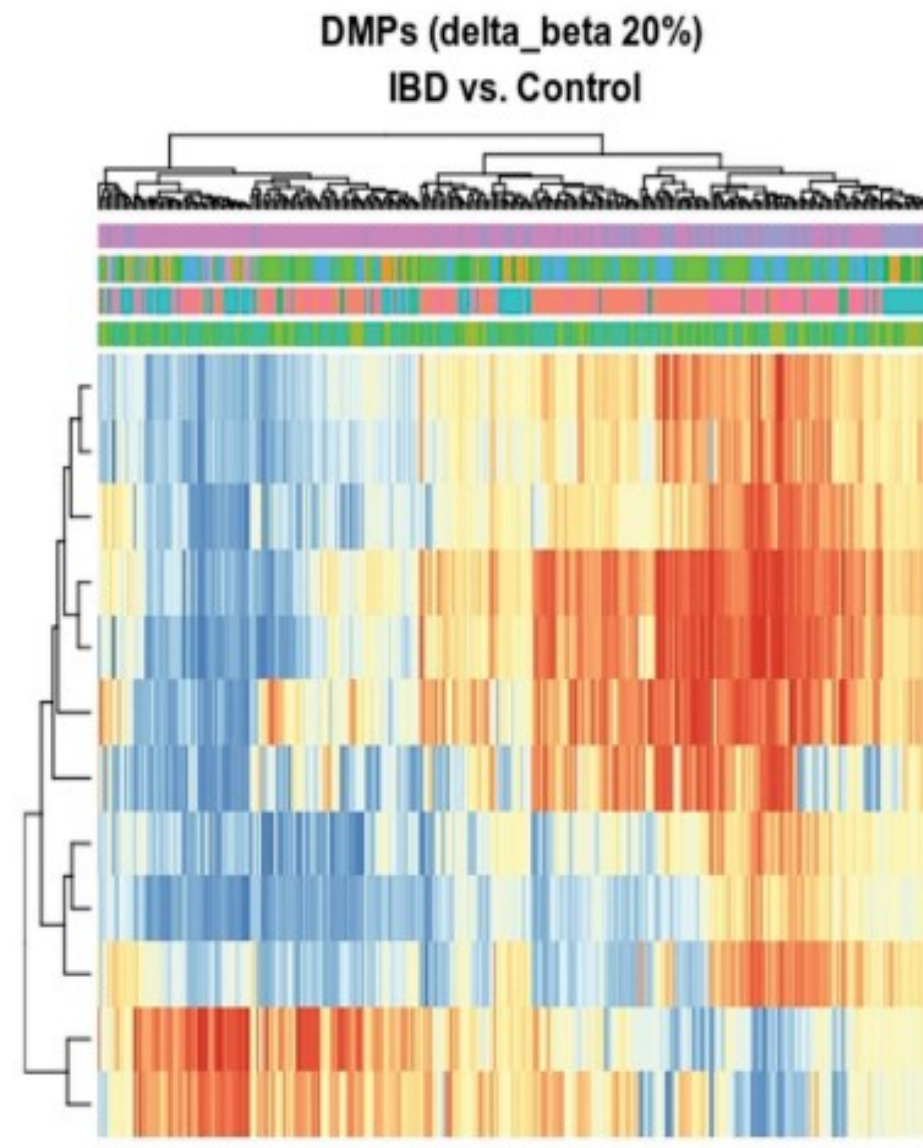
In UC, an epigenetic imbalance occurs



DNA methylation influences inflammation by modulating the expression of key genes in the immune response - homeostasis and defense

Methylation reported in IBD

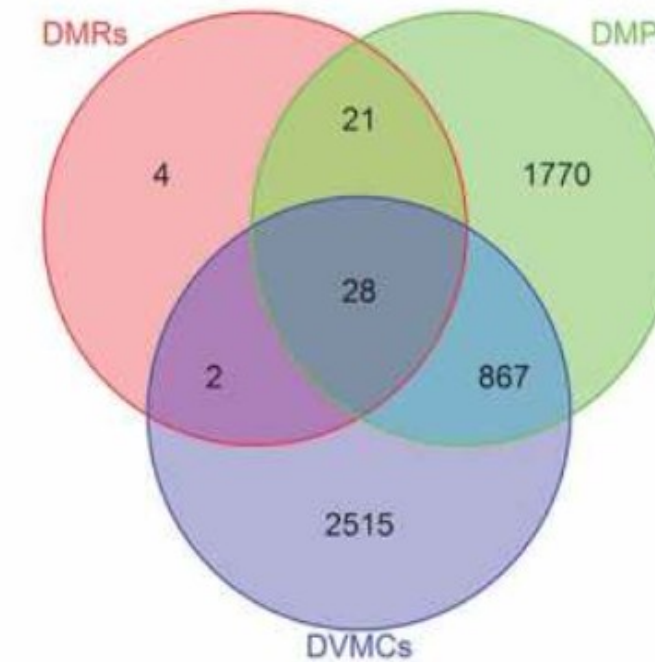
Abnormal DNA methylation in human inflammatory conditions of the gastrointestinal tract - IBD



Heat map showing differentially methylated positions between IBD and controls.

DNA methylation may be both the consequence (and the potential effector) of genetic susceptibility variants as well as environmental signals and cytokine exposure

Overlap between DMPs, DMRs, and DVMCs
overlapping gene symbols



- **DMRs (rojo):**
 - Regiones diferencialmente metiladas
 - Metilación promedio en regiones
- **DMPs (verde):**
 - Sitios CpG diferencialmente metilados
 - Metilación promedio en sitios individuales
- **DVMCs (azul/morado):**
 - Sitios CpG diferencialmente variables y metilados

Suggests that key genes are altered in multiple ways, highlighting their potential as disease biomarkers.

Pathway

Adipogenesis genes_Mus musculus_WP447

Adipogenesis genes_Homo sapiens_WP236

TGF Beta Signalling Pathway_Mus musculus_WP113

TGF-beta Receptor Signalling_Homo sapiens_WP560

Alpha6-Beta4 Integrin Signalling Pathway_Mus musculus_WP488

Haemostasis_Homo sapiens_R-HSA-109582

G alpha(12/13) signalling events_Homo sapiens_R-HSA-416482

Pathways in cancer_Homo sapiens_hsa 05200

Aldosterone synthesis and secretion_Homo sapiens_hsa04925

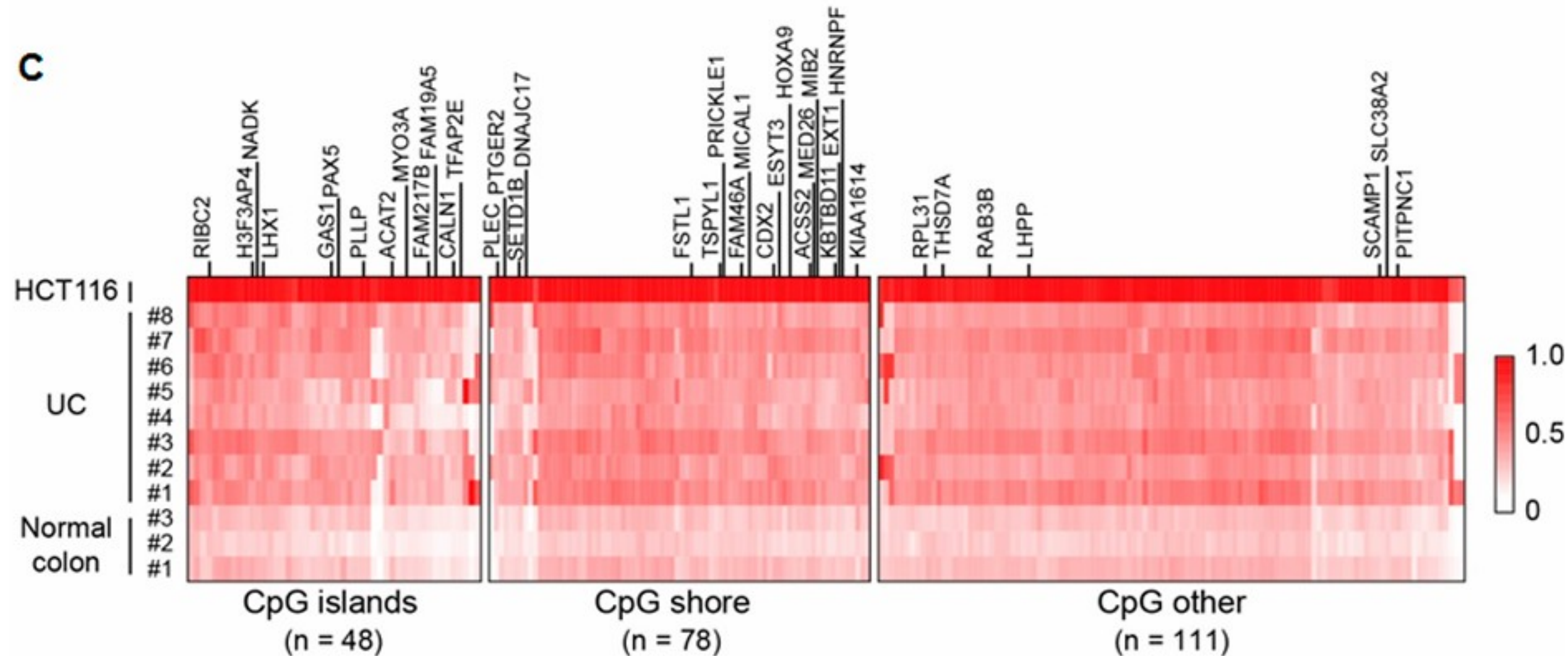
Chronic inflammation

Loss of intestinal epithelial integrity

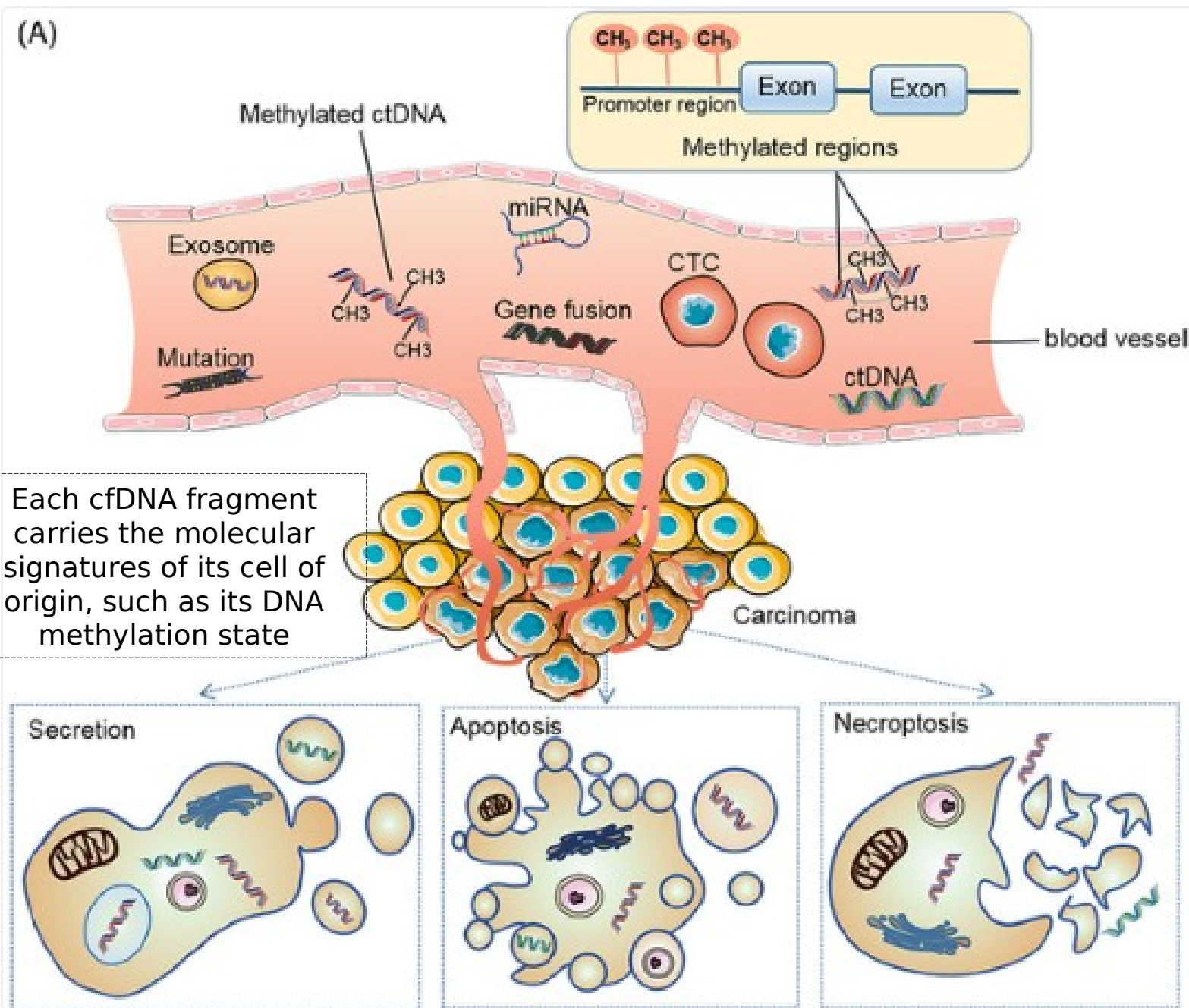
Cancer precursors with epigenetic changes in IBD

Methylation GWAS reported in UC

The cause of IBD remains unknown, but there is growing evidence that environmental factors such as epigenetic changes may contribute to the etiology of the disease.



UC patients show an altered methylation profile that differs from normal colon and partially resembles that observed in colorectal cancer cells.
This supports the idea that epigenetic changes in UC may contribute to increased cancer risk.

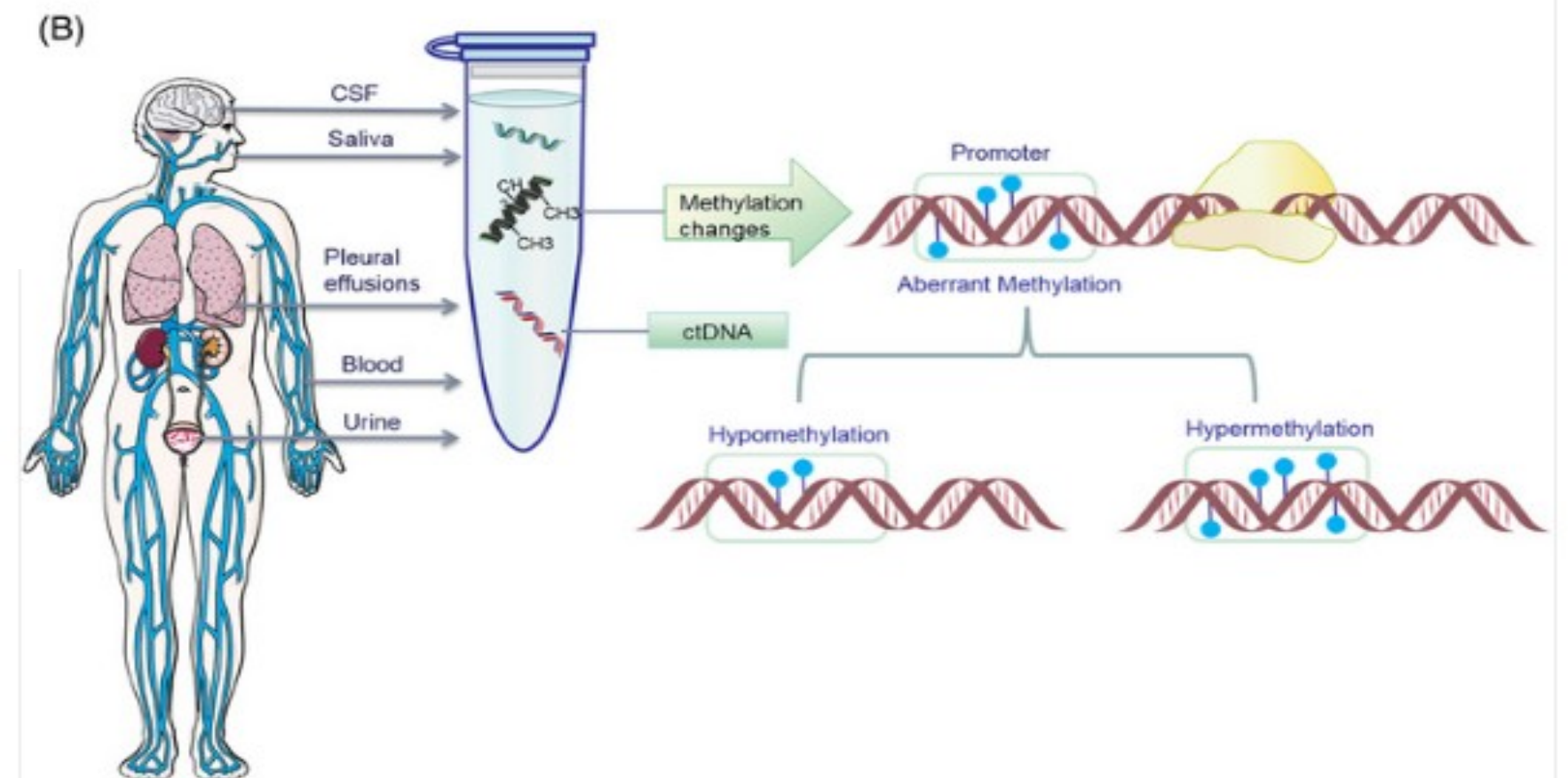


This aligns with the interest in studying epigenetic signatures and their usefulness in risk stratification in UC patients at high risk of developing CRC.

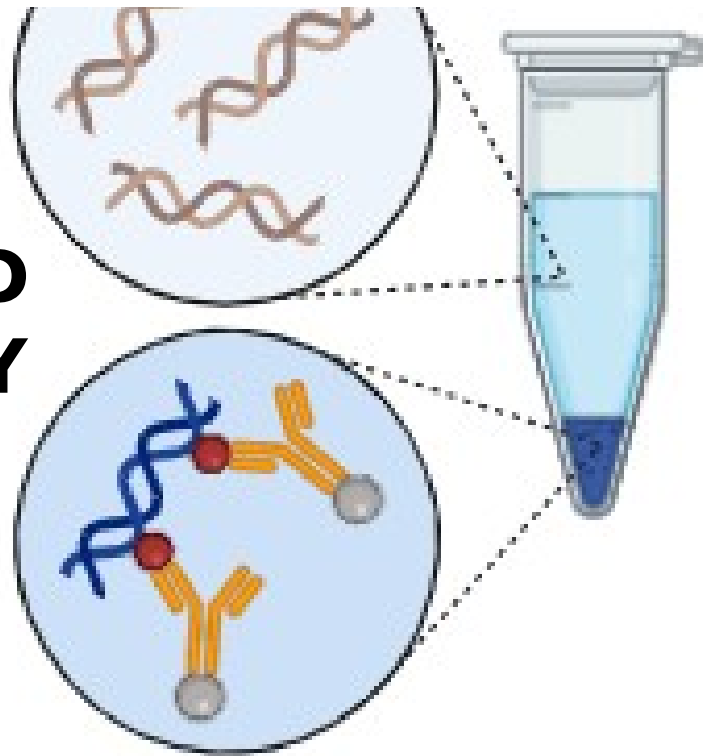
Free plasma DNA

The ctDNA methylation profile is a **FUNDAMENTAL** tool for:

- ✓ Early detection
- ✓ Recurrence monitoring
- ✓ Prognostic prediction of diseases that may develop into various types of cancer



LIQUID BIOPSY

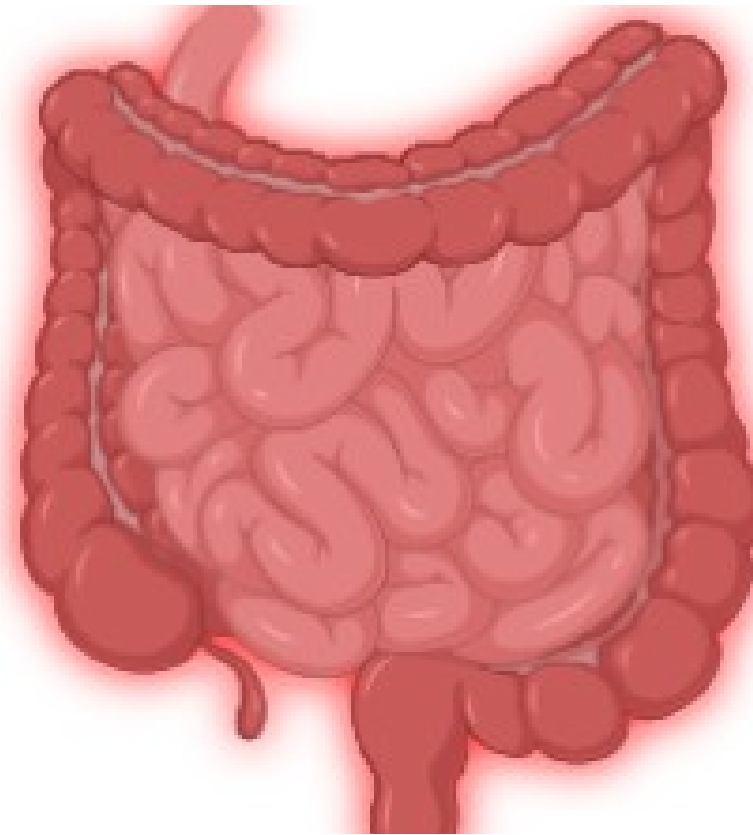


Serological and DNA methylation biomarkers

Offer a promising alternative, as they are less invasive and may better reflect the underlying biological mechanisms, such as the epigenetic changes that may contribute to inflammation and carcinogenesis associated with UC.

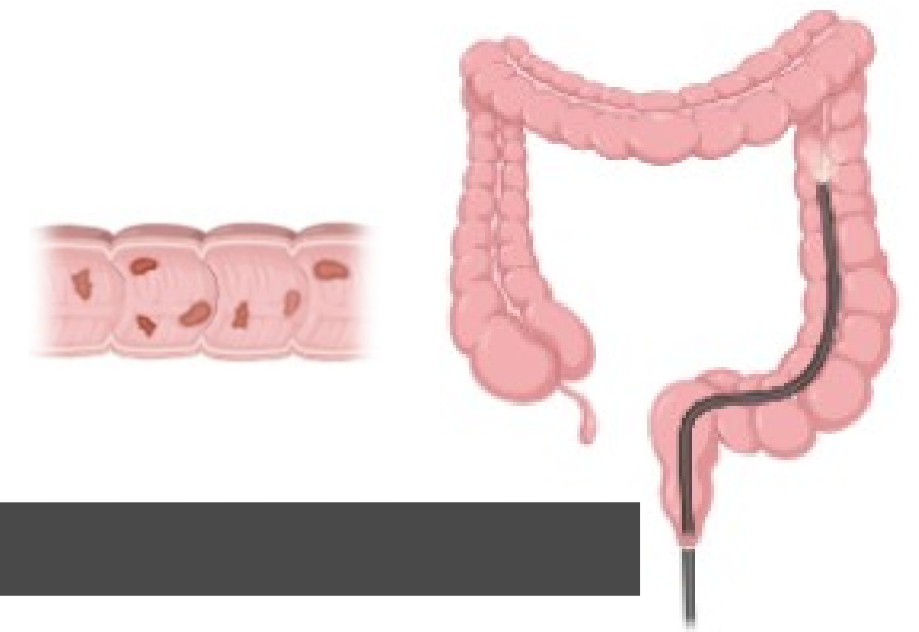
UC is a chronic inflammation of the intestinal mucosa.

It has been shown that patients with long-standing disease and/or extensive involvement have a higher risk of developing CRC



Problem Statement

Many diagnostic methods are invasive and lack precision to predict progression to CRC



RESEARCH QUESTION



1. Is there a differential methylation signature in cfDNA in UC patients associated with CRC?
2. Can SIgA levels be used as a prognostic biomarker in patients with ulcerative colitis (UC)?

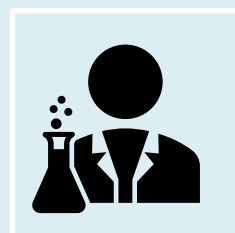
HYPOTHESIS

- In patients with UC at high risk of developing CRC, differentially methylated signatures exist in cfDNA and colonic tissue compared to what is observed in healthy individuals - CRC
- In UC patients, SIgA levels are associated with the methylation signature in UC patients at high risk of developing CRC



Rationale

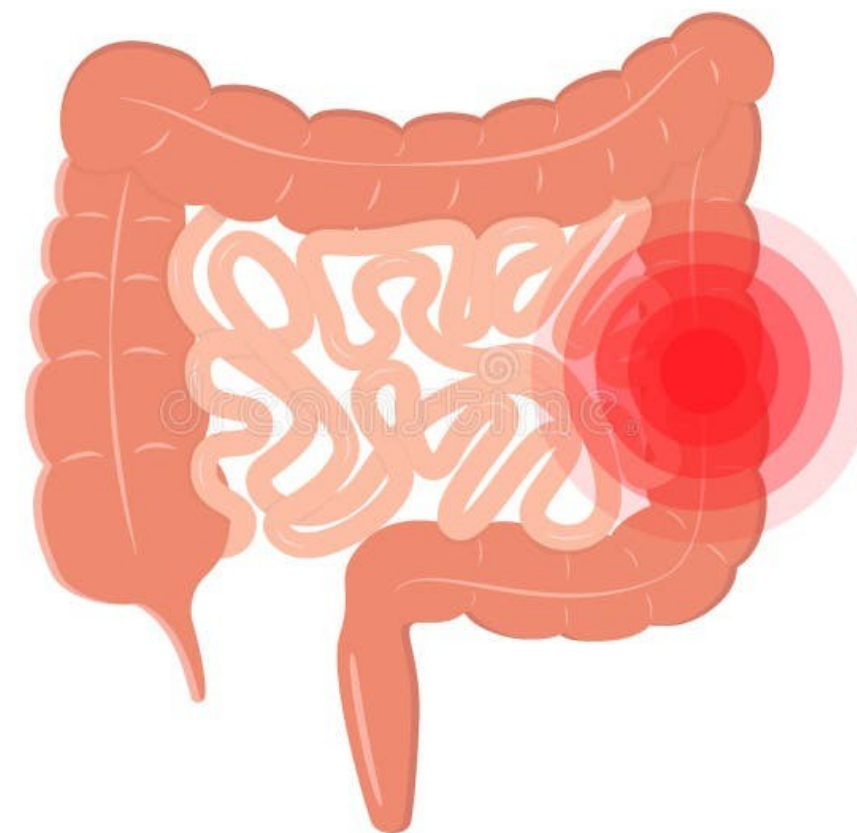
Limited research on cfDNA methylation patterns in the plasma of patients with severe UC and their association with the regulation of SIgA production



We propose to analyze a differentially methylated gene signature between healthy individuals and patients with long-standing UC, which could serve as biomarkers to identify those at greater risk of progressing to CRC and to associate them with SIgA production levels.



By implementing liquid biopsy as a diagnostic, minimally invasive tool, we can contribute to the early identification of epigenetic alterations associated with the disease



“Integrating both dimensions could open new possibilities for identifying combined biomarkers with diagnostic and prognostic value in UC patients at high risk of developing CRC”

General Objective

To establish the association between the methylation signature in liquid biopsy and colonic tissue with SIgA levels in UC patients at high risk of developing CRC

SPECIFIC OBJECTIVES

- 1. Characterize the free DNA methylation profiles in plasma and colon biopsies of UC patients at high risk of colorectal cancer.**
2. Compare the free DNA methylation profiles in plasma and colon biopsies of UC patients versus healthy controls and CRC patients
3. Associate serum SIgA levels with the methylation signature identified in liquid biopsies and colonic tissue of UC patients presenting symptoms of high colorectal cancer risk.
- 4. Analyze the associations between methylation patterns, secretory IgA levels, and inflammation markers in UC patients at high risk of CRC.**

Methodology

Design

*Descriptive, cross-sectional study
with an analytical component*



Study population:

Patients with confirmed UC.

Controls: CRC and healthy.

Centers: Cobos Medical Center, HIC Bucaramanga,

GastroAdvanced.

Gastroenterologists: Viviana Parra – Cristian Flores –

Ricardo Villareal

Samples (36 total)

UC: 8 circulating DNA + 8 colonic tissue.

CRC: 5 circulating DNA + 5 colonic tissue

Healthy: 5 colonic tissue + 5 circulating DNA



Inclusion criteria for UC:

- Patients with UC > 8 years or with sclerosing cholangitis.

Exclusion criteria for UC:

- No Crohn's disease
- Contraindication to colonoscopy/biopsy
- Recent prior colonic surgery for UC.

Methodology for objective 1. Characterize the free DNA methylation profiles in plasma and colon biopsies of UC patients at high risk of colorectal cancer

Methodology for objective 2. Compare the free DNA methylation profiles in plasma and colon biopsies of UC patients versus healthy controls and CRC patients

1. Preparation of colonic tissue and blood plasma samples for DNA extraction:

- Colonic tissue biopsies from patients with confirmed UC, CRC patients, and healthy controls. This will be performed under the supervision of a gastroenterologist, through biopsy, colonoscopy, and chromoendoscopy
- Tissue samples will undergo a fragmentation process using the NEBNext UltraShear kit



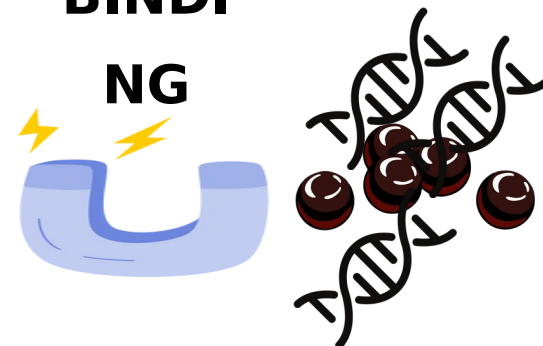
2. Blood plasma samples will be collected in EDTA tubes; blood will be centrifuged at 3600  min to obtain plasma

LYSIS



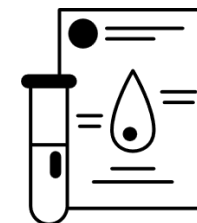
Proteinase K and lysis buffer

BINDING



cfDNA binds to the magnetic beads.

WASH



Removal of impurities through automated washes.

ELUTION



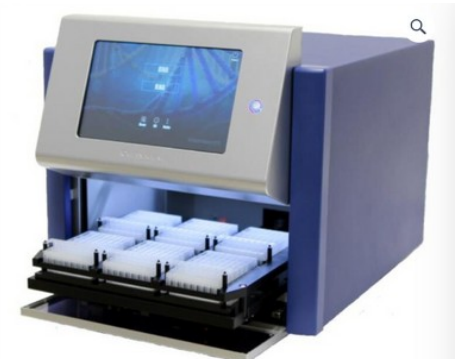
Recovery of purified cfDNA in elution buffer.

QUANTIFICATION



Quantification of nucleic acids ng/μl

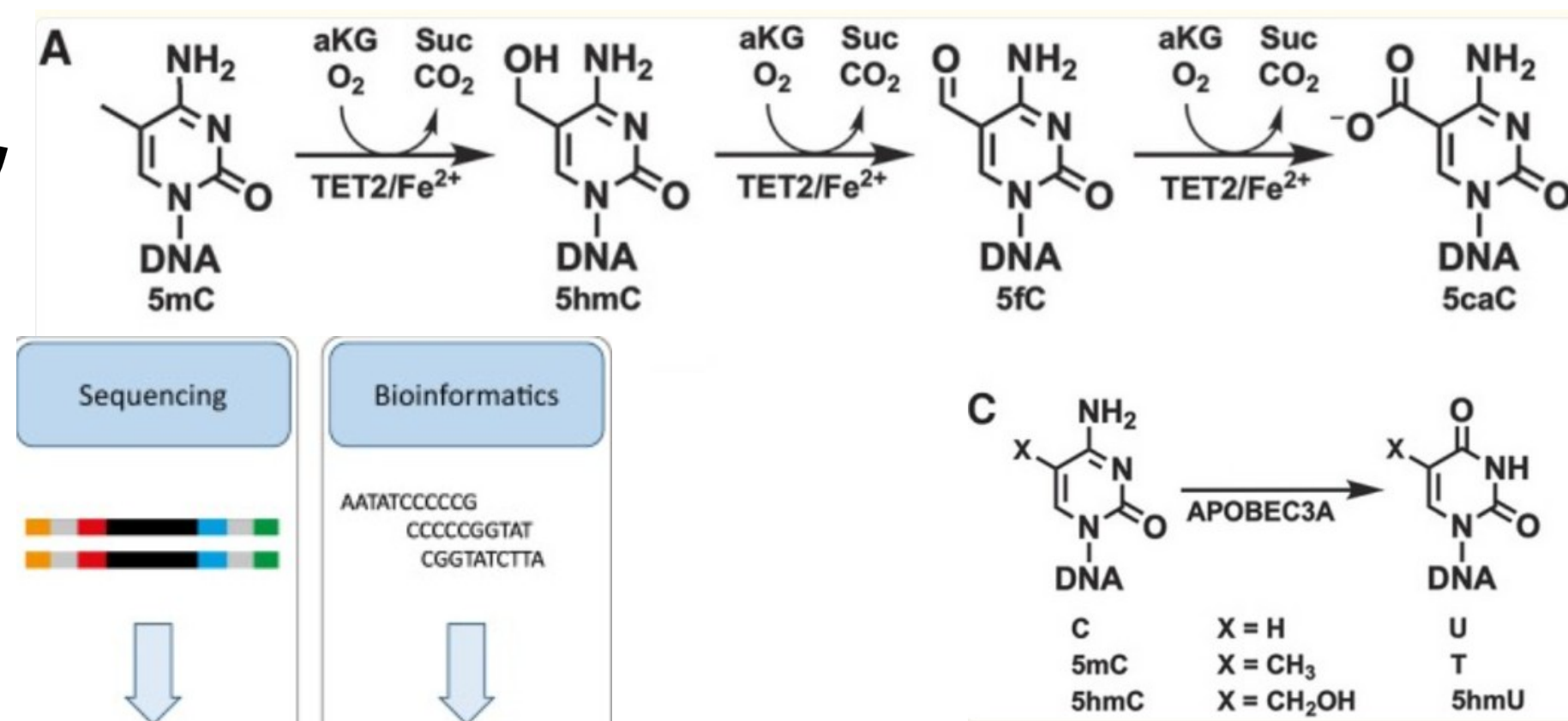
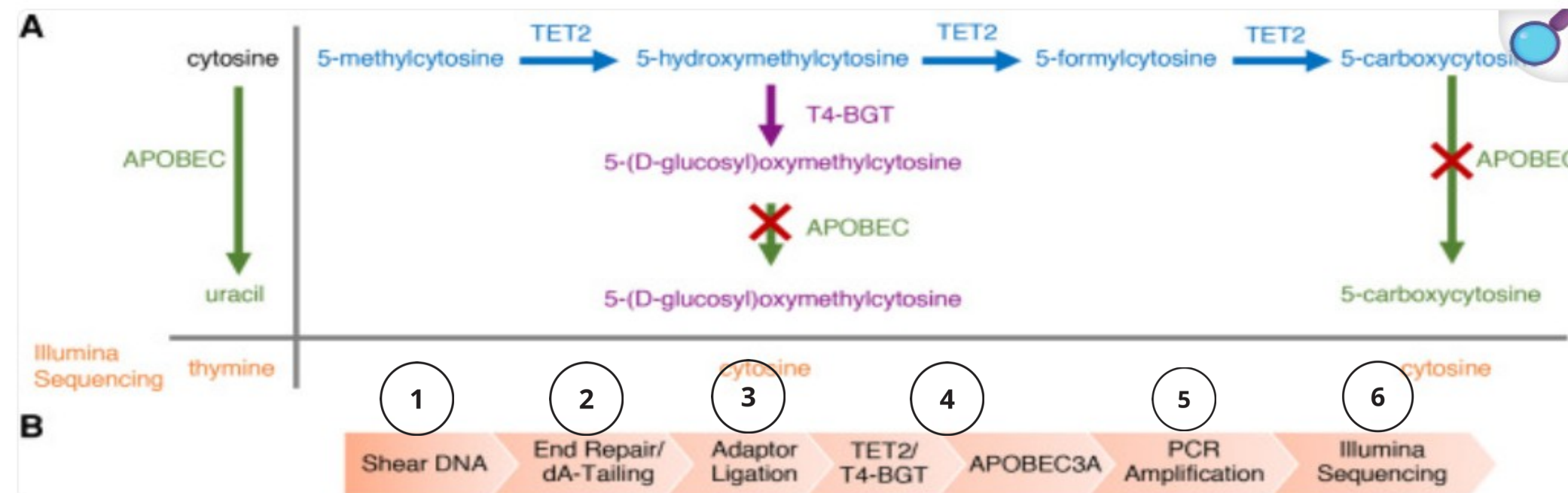
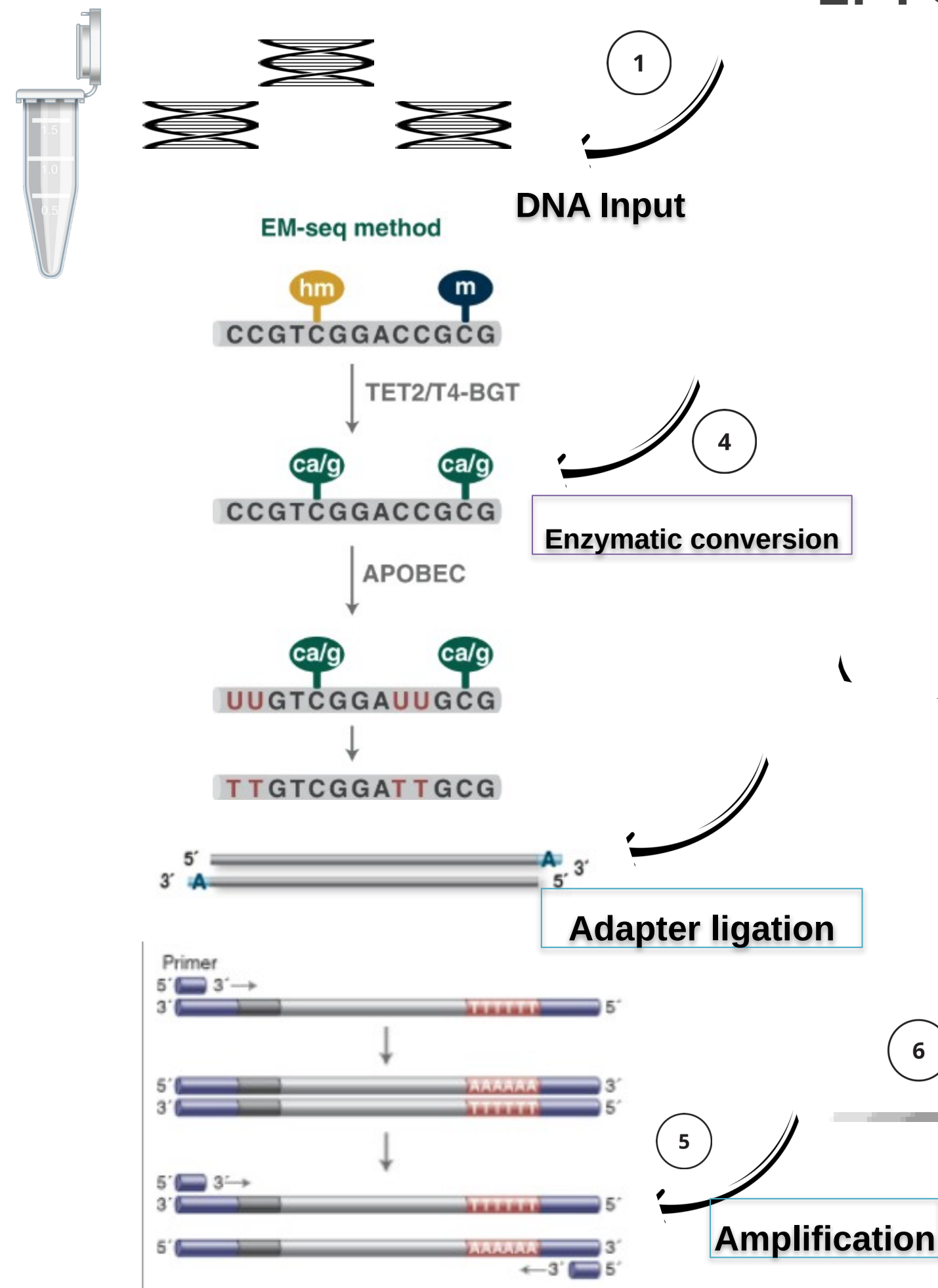
PANAMAX 48



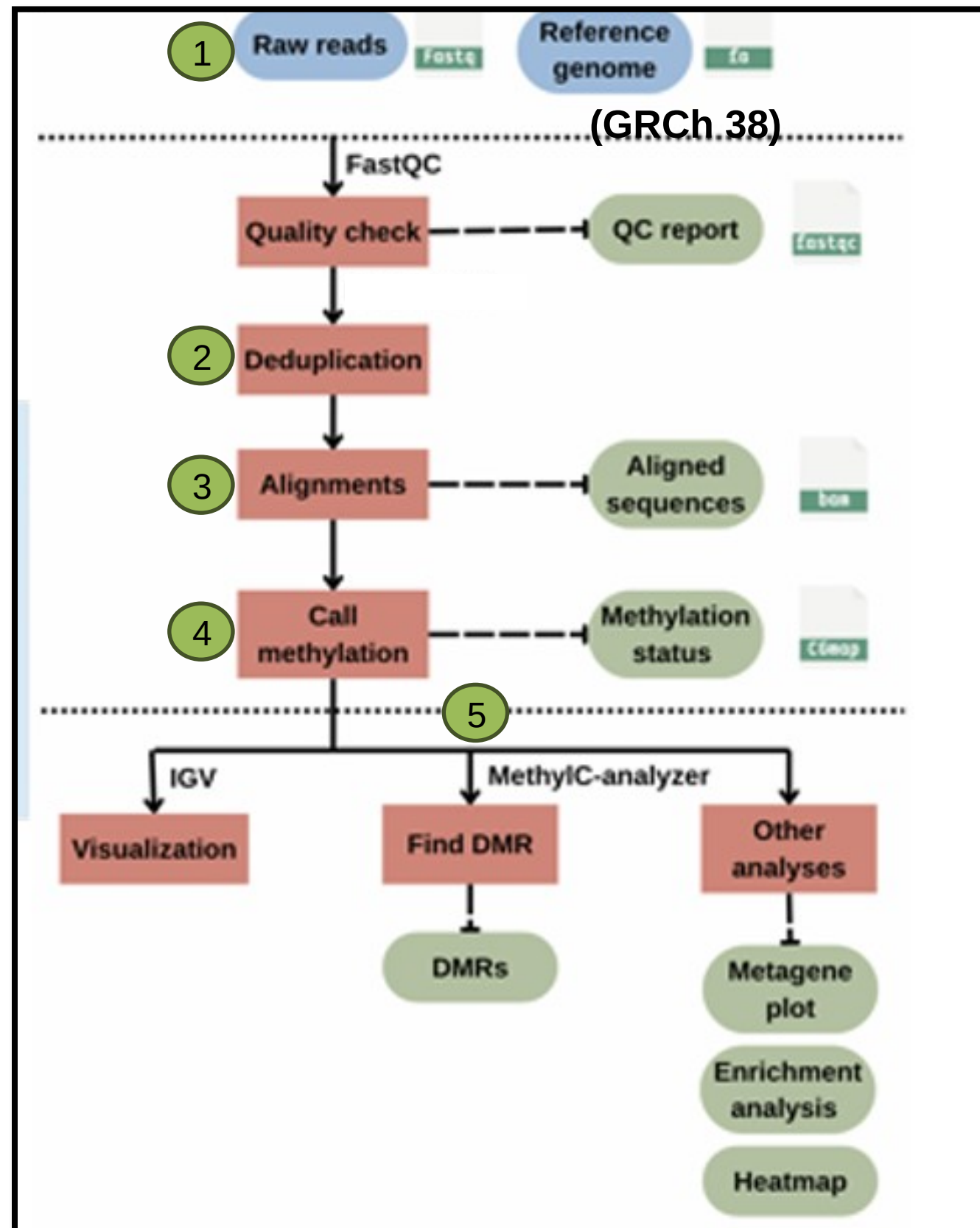
3. Tissue purification - cfDNA with magnetic beads (PANAMAX™ 48).

Steps: Lysis → Binding → Wash → Elution

EM-SEQ Library Preparation



Bioinformatic Analysis



- 1 FastQC: FASTQ → QC report.
- 2 Deduplication: FASTQ → cleaned FASTQ.
- 3 Alignment (BS-Seeker2/bowtie2): FASTQ+FASTA → BAM.
- 4 Methylation calling: BAM → CGmap (+ optional
- 5 IGV: WIG/BigWig/BED/GTF → visualization.
 - MethylC-analyzer: CGmap+GTF → DMRs, metagene, enrichment, heatmaps.

Comparison of Epigenetic Signatures

Regression models to assess significant differences between groups. A DSS (Dispersion Shrinkage for Sequencing data) analysis available in RStudio will be performed to identify CpG regions between plasma vs. colonic tissue

DSS: uses a beta-binomial model with dispersion shrinkage to test methylation differences between groups and/or conditions

The proportion of 5mC will be calculated using the following formula:

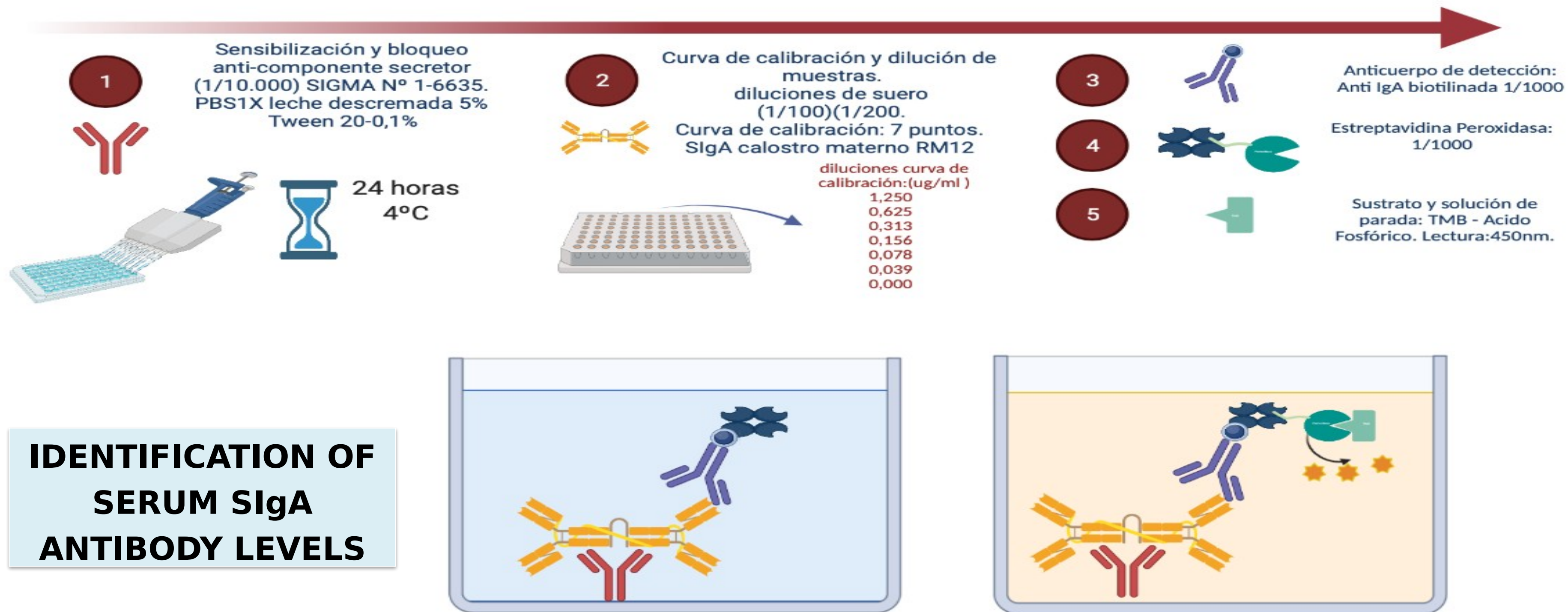
$$\% \text{ methylation} = \frac{\# \text{ methylated cytosines}}{\# \text{ methylated cytosines} + \# \text{ unmethylated cytosines}} \times 100$$

This proportion will be compared in paired cfDNA and colonic tissue samples from 5mC counts per CpG site using the paired Wilcoxon test.

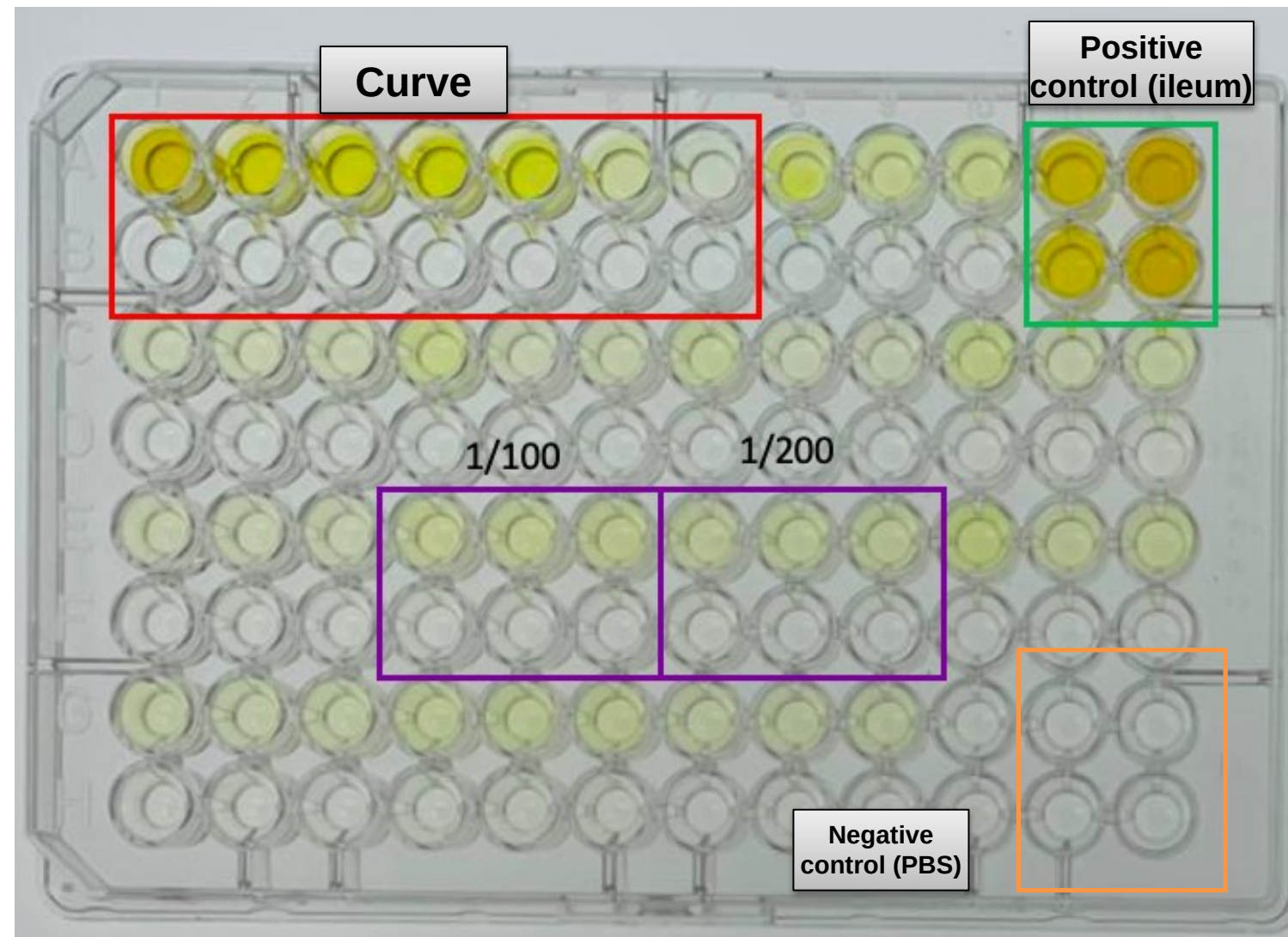
KEGG gene enrichment analysis to identify biological pathways impacted by the epigenetic changes.



Methodology for objective 3. Associate serum SIgA levels with the methylation signature identified in liquid biopsies and colonic tissue of UC patients presenting symptoms of high colorectal cancer risk



Methodology for objective 3. Associate serum SIgA levels with the methylation signature identified in liquid biopsies and colonic tissue of UC patients presenting symptoms of high colorectal cancer risk



Reading at 450 nm

Optical density (OD) is measured at 450 nm on the reader.

Normalize against the blank

The blank is the well with no sample.

Normalize = subtract each well's OD from the blank's OD

$$\text{Corrected OD} = \text{Sample OD} - \text{Blank OD}$$

Validity criteria and control

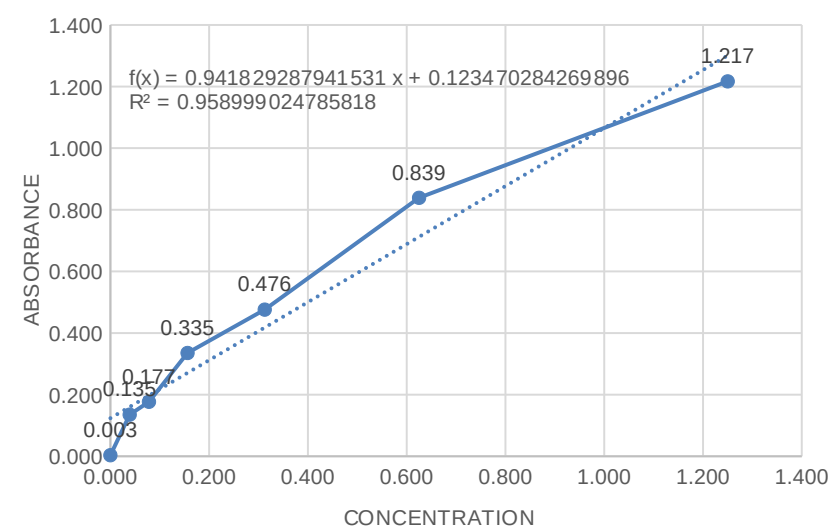
Positive control (colon 1:100 or ileum): Must give a high absorbance (≥ 0.2 and at least double the blank).

Negative control (PBS): Very low absorbance, close to the blank.

Standard curve → SIgA concentrations

Samples will be compared with the standard curve to obtain SIgA concentrations.

Standard curve



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Methodology for objective 3. Associate serum SIgA levels with the methylation signature identified in liquid biopsies and colonic tissue of UC patients presenting symptoms of high colorectal cancer risk

Methodology for objective 4: Analyze the associations between methylation patterns, secretory IgA levels, and inflammation markers in UC patients at high risk of CRC



Recording and software

Results were recorded in Excel and analyzed in SPSS v26 (Windows).

Descriptive analysis

Continuous variables: measures of central tendency.

Categorical variables: frequency analysis.

Inferential analysis

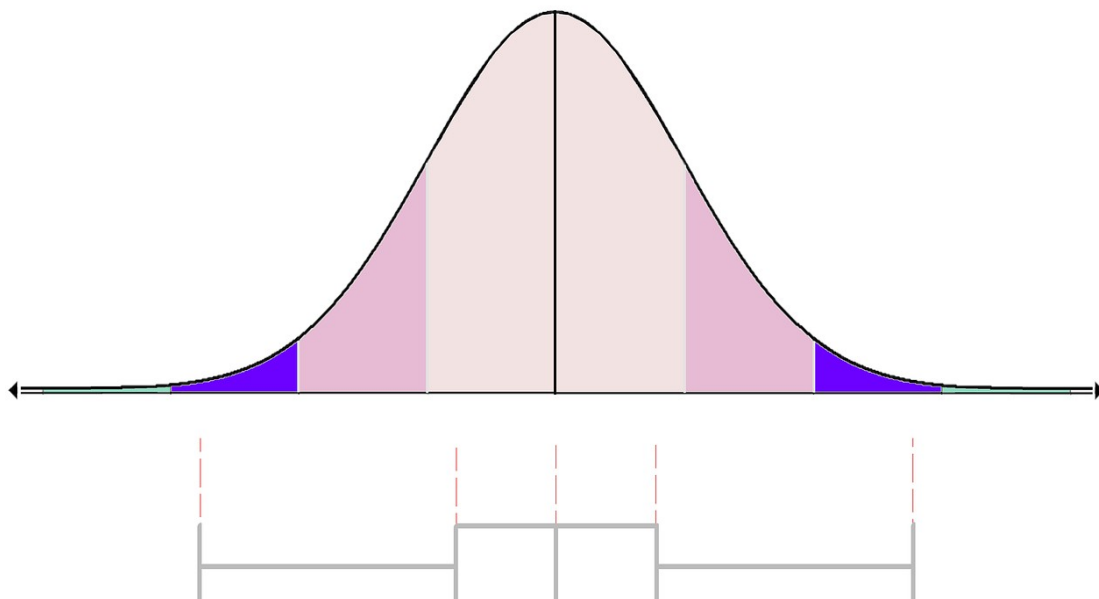
Relationship between variables: Chi-square test or Fisher's exact test, depending on statistical assumptions.

Significance criteria

Confidence level: 95%.
p-values ≤ 0.05 considered significant

A descriptive statistical analysis will be performed on demographic variables such as

- age
- sex
- origin
- family history of cancer
- cigarette use
- disease duration
- presence of sclerosing cholangitis



Ethical Considerations

- Colombian regulation established in Resolution 8430 of 1993, which establishes the scientific, technical, and administrative standards for health research.
- This research will take into account the articles referenced in Chapter I of said resolution concerning the ethical aspects of research in humans, and given the characteristics of the study, it is classified as greater than minimal risk (colonoscopy is clinically indicated); every participant will sign an informed consent form before starting the study.
- This research project has ethical approval from the research committee of HIC Bucaramanga - Fundación Cardiovascular de Bucaramanga, Colombia.
- **Minutes No. 005-2024**



Expected Results

1

cfDNA methylation profiles and biopsies from UC patients are expected to be characterized, revealing specific methylation patterns that may serve as early diagnostic biomarkers. This will make it possible to identify significant differences compared with healthy individuals

2

The correlation between SIgA levels and methylation patterns is expected to show that certain genes related to immune response regulation are hypermethylated, suggesting that DNA methylation may affect SIgA production

3

When evaluating SIgA levels in patients with the disease and their relationship with methylation signatures, these levels are expected to be decreased in UC and CRC patients compared to healthy individuals, which could indicate dysfunction in the local immune response.

4

Finally, by analyzing all of the factors described, these will provide a deeper understanding of the underlying pathogenic mechanisms and their potential application in CRC surveillance and prevention strategies.



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Budget

PCI-2024-0016

	VALOR EN MILES DE PESOS	JUSTIFICACIÓN	ENTIDAD 1	ENTIDAD 2	TOTAL, EN MILES DE PESOS
PERSONAL	\$ 85.938.228				\$ 85.938.228
MATERIALES	\$ 49.950.000	Material de toma y almacenamiento de las muestras de sangre. Material de toma y almacenamiento de las muestras de tejido. Material/Reactivos. Extracción ADN de tejido y sangre periférica Material/Reactivos. Medición IgA secretora. Material procesamiento de muestras	UEB		\$ 60.201.574

EQUIPOS NUEVOS	0				0
EQUIPOS DE USO PROPIO	\$ 95.928.000				\$ 95.928.000
SERVICIOS TECNICOS	\$ 27.000.000	Secuenciación de metiloma			\$ 27.000.000
VIAJES	\$ 7.000.000	Difusión de resultados en evento científico			\$ 7.000.000
PUBLICACIONES	\$ 4.000.000	Publicación de artículo científico			\$ 4.000.000
TOTAL					\$ 280.067.802

